MIL-B-21442B(Wep)
13 August 1963
Superseding
MIL-B-21442A(Wep)
17 February 1960

MILITARY SPECIFICATION

BATTERY, STORAGE
NICKEL-CADMIUM

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope. This specification states the requirements for the procurement of a battery, Bureau of Naval Weapons Drawing 1413631, which is used to provide electrical energy. The battery is of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	Plastic Film (Polyethlene Thin Gage)

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-901	Shockproof Equipment, Class HI (Hi Impact), Shipboard Application, Tests for
MIL-B-22205	Bags, Transparent, Flexible, Heat Sealable, for Packaging Applications

FSC 6140

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

NAVY DEPARTMENT

BUWEPS ID 495960	Shipping (Packing) Condition (For 6 Batteries)
BUWEPS Drawing 1413631	Battery, Storage

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

OS 6341	Ordnance Design Requirements, Miscellaneous General

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw-Threads Standards for Federal Service

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Description. The battery is a sintered-plate, nickel-cadmium storage battery that will furnish an average of 4.15 plus or minus 0.35 amperes of current at 12.5 plus or minus 1 volts for a minimum of 8 minutes.

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3.2 General requirements

3.2.1 Preproduction samples. Unless otherwise specified in contract or order, preproduction samples of the battery shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification and the design. See 4.1.2.

3.2.2 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the battery shall be selected in accordance with BUWEPS Drawing 1413631.

3.2.3 Surface treatment, construction, and dimensions. Surface treatment, construction, and dimensions for the battery shall be in accordance with the applicable drawings, specifications, and publications listed in BUWEPS Drawing 1413631.

3.2.4 Moisture, fungus, and corrosion. The selection of materials and protective coatings for all exposed surfaces of the battery shall be in accordance with OS 6341.

3.2.5 Marking. Each battery shall contain an identification plate or marking containing the following:

(a) Manufacturer's name and location.

(b) Name of item and BUWEPS drawing number.

(c) Manufacturer's catalog number and serial number.

(d) Contract number.

3.2.6 Gages. The contractor shall provide himself with whatever gages are necessary and adequate to insure that the materials to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

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3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Maintenance instructions. Maintenance instructions, including charging instructions, shall accompany each battery.

3.2.9 One syringe for adjusting the electrolyte level shall be included for every two batteries.

3.2.10 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.3 Definitions. Definitions applicable to the environmental requirements of this specification are as follows:

3.3.1 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock, in general, can be described as a buildup of 90 percent of peak force in less than 0.010 second. Shock is described (a) as a pulse in gravity (g), duration in milliseconds (ms) and some description of the shock wave form, or (b) in terms of test conditions outlined in Specification MIL-S-901.

3.3.2 Battery axes. The axes of the battery are:

(a) Longitudinal

(b) Perpendicular and transverse to (a)

(c) Mutually perpendicular to (a) and (b)

3.4 Performance requirements and product characteristics. The battery shall meet the following performance requirements and product characteristics.

3.4.1 Detail performance characteristics

3.4.1.1 Output voltage

3.4.1.1.1 Initial. From 15 seconds through 6 minutes after start of discharge, the output voltage shall be 12 plus or minus 1 volts while delivering output current as stated in 3.4.1.2.1.

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3.4.1.1.2 Final. At the end of 8 plus 1 minus 0 minutes the output voltage shall be a minimum of 10.5 volts while delivering the minimum output current, as stated in 3.4.1.2.2.

3.4.1.2 Output current

3.4.1.2.1 Initial. From 15 seconds through 6 minutes after start of discharge, the output current shall be 4 plus or minus 0.33 amperes.

3.4.1.2.2 Final. At the end of 8 plus 1 minus 0 minutes the output current shall be a minimum of 3.5 amperes.

3.4.2 Detail product characteristics

3.4.2.1 Vents. Each cell shall be equipped with a vent that will permit the escape of gas during charge.

3.4.2.2 The battery shall be capable of receiving a full charge at a constant potential of 15 volts with conditions as specified in 4.4.2 within 1.5 hours. The battery shall also be capable of receiving a full charge when a constant current of 0.875 plus or minus 0.125 ampere is applied for a period of 4 hours maximum.

3.4.2.3 Delivery condition. The battery shall be delivered complete with electrolyte and in the discharged condition.

3.5 Environmental requirements. The battery shall be capable of withstanding the environmental requirements and tests specified in this specification and thereafter meet all the performance requirements and product characteristics specified herein.

3.5.1 Shock. The battery shall withstand a shock of 60 g peak magnitude of not less than 8 ms duration, and with a pulse approximating a one-half sine wave, and shall function before and after the shock test. Thereafter the battery shall show no damage and shall meet all performance requirements and product characteristics of this specification.

3.5.2 Shock; shipping condition. The battery packaged and packed as specified in section 5 shall withstand the rough handling test of MIL-P-116 and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.3 Temperature. The battery shall meet the performance requirements and product characteristics of this specification while operating at temperatures between zero degrees Fahrenheit (F) and plus 130 degrees F (minus 18 degrees Centigrade (C) to plus 54 degrees C).

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3.5.4 Humidity. Up to 95 percent of relative maximum.

3.5.5 Life. The battery shall be capable of 100 cycles of charge and discharge in accordance with 4.5.5.

3.5.6 Standtime. The battery shall meet the test of 4.4.4.4 after having been charged as in 4.4.4.3 and then left unused for a minimum of 60 days.

3.5.7 Inversion. There shall be no evidence of electrolyte leakage when the battery is turned bottom-side up and tested in accordance with 4.5.1.

3.6 Cleanup. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of excess electrolyte and other foreign matter.

3.7 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of marking of parts and assemblies, plating, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.2 Preproduction sample. A sample of five batteries shall be submitted to the Government inspector for preproduction tests at an activity designated in the contract or by the Bureau of Naval Weapons for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity on the preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.3 Lot size. Unless otherwise specified, the number of units of product in "inspection lots" will be as determined by the Government inspector.

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4.1.4 Periodic production samples. From each succeeding lot or grand lot of 300 or less, a random sample of 5 batteries will be selected by the Government inspector for submission to a Bureau of Naval Weapons assigned activity to determine compliance of the samples with the requirements of the contract, specifications, and drawings. The battery shall be considered satisfactory if it meets all specifications and pertinent drawing requirements and withstands the tests specified in 4.4 and 4.5 inclusive. Failure to meet these requirements may result in termination of contract requisition or order for default.

4.2 Classification of test. The inspection and testing of the battery shall be classified as follows:

4.2.1 Acceptance tests. Acceptance tests are to be accomplished on each battery being submitted for acceptance under contract. These tests, detailed in 4.4, shall be performed by the contractor.

4.2.2 Preproduction and periodic tests. Preproduction and periodic production tests are those which are accomplished on samples selected as specified in 4.1.2 and 4.1.4, which are representative of the production of the battery after the award of the contract, to determine that the production meets the requirements of this specification. These tests are detailed in 4.5. Acceptance shall be based on no defects in the sample. Failure of the sample to comply with these requirements will result in the rejection of the lot or the cessation of production, as determined by the procuring activity.

4.3 Test equipment. The following items of test equipment are required to perform the acceptance tests set forth in this specification:

4.3.1 Battery charger. Capable of supplying a constant voltage of 15 volts at 15 amperes or a constant current of 1 ampere with a potential capable of varying up to approximately 18 volts.

4.3.2 Voltmeter. Zero to 20 volts direct current (dc) voltmeter with an accuracy of plus or minus 2 percent.

4.3.3 Ammeter. Zero to five ampere dc ammeter with an accuracy of plus or minus two percent.

4.3.4 Load resistor. Load resistor rated 3 ohms plus or minus 1 percent and 150 watts.

4.3.5 Time clock. Zero to 15 minutes with an accuracy of plus or minus 2 percent.

4.3.6 Switch. Single-pole, single-throw capable of carrying 5 amperes of direct current with a maximum voltage of 20 volts.

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4.4 Acceptance tests

4.4.1 The contractor shall conduct acceptance tests as specified in 4.2.1 to ensure that the battery is in compliance with the requirements of 3.2, 3.4, 3.6, and 3.7.

4.4.2 Test conditions. Unless otherwise specified, the battery shall be subjected to acceptance tests under the following conditions.

4.4.2.1 Temperature. Room ambient 65 degrees F to 95 degrees F (18 degrees C to 35 degrees C).

4.4.2.2 Altitude. Normal ground atmospheric pressure.

4.4.2 3 Vibration. None.

4.4.2.4 Humidity. Room ambient to 95 percent relative maximum.

4.4.3 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.4 Test procedure

4.4.4.1 Delivery condition. The battery shall be delivered for acceptance tests complete with electrolyte and in the discharged condition.

4.4.4.2 Preparation. The battery shall be allowed to stand in open circuit for not less than two hours until battery and electrolyte reach standard conditions (see 4.4.2) or a temperature not more than ten degrees F above ambient temperature.

4.4.4.3 Charging. After its preparation charge the battery (4.4.4.2) using the constant voltage method (3.4.2.2) and battery charger 4.3.1. The cell covers shall allow the escape of gas during the charging procedure.

4.4.4.4 Discharge test

4.4.4.4.1 The battery shall be stored at an ambient temperature in accordance with 4.4.2 until the battery reaches ambient temperature.

4.4.4.4.2 Connect in a series circuit the battery, a switch (4.3.6), an ammeter (4.3.3), and load resistor (4.3.4). Connect the voltmeter (4.3.2) across the load resistor.

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4.4.4.4.3 Close the switch. From 15 seconds through 6 minutes after start of discharge, the battery voltage and output current shall be as specified in 3.4.1.1.1

4.4.4.4.4 At the end of the eight plus one minus zero minutes, the battery voltage and output current shall be as specified in 3.4.1.1.2.

4.4.4.5 Cycling test. Repeat 4.4.4.3 and 4.4.4.4 twice.

4.5 Preproduction and periodic production tests

4.5.1 Inversion. The battery, when filled to the normal level (red line) with electrolyte and with the cell vents secured in position, shall, within 2 to 24 hours after being charged, be turned bottom-side up and subjected to the following vibration schedule without evidence of electrolyte leakage:

Frequency Range	Amplitude	Time	Temperature
10 - 30 cps	0.060 in. ± 0.006 in.	30 min.	Room
31 - 60 cps	3 g ± 0.3 g	30 min.	Room
61 - 100 cps	3 g ± 0.3 g	30 min.	Room

4.5.2 Shock. The battery shall be subjected to shock tests so as to meet the requirements specified in 3.5.1 and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.2.1 Mounting. The battery shall be mounted securely on its base without cushioning and shall be rigidly clamped to prevent its movement.

4.5.2.2 Procedure. The battery shall be subjected to 3 shock pulses of 60 g plus or minus 6 g peak magnitude of not less than 8 ms pulse duration and approximating a one-half sine wave in each of the 3 axes defined in 3.3.2.

4.5.3 Shock; transportation, storage, and handling. The battery, packaged and packed as specified in section 5, shall be subjected to the rough handling tests for packs not exceeding 200 pounds set forth in Specification MIL-P-116.

4.5.4 Temperature. The battery shall be subjected to temperature tests so as to meet the requirements specified in 3.5.3 and thereafter shall meet the performance requirements and product characteristics of this specification.

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4.5.4.1 Low temperature operation. The battery shall be placed within a test chamber. The test chamber temperature shall be lowered to zero degrees F plus or minus 5 degrees F and maintained at this temperature for a period of 48 hours plus or minus 5 hours. At the end of this test period the battery, while still at the test temperature of zero degrees F, shall be charged, using the constant current method (3.4.2.2) and battery charger (4.3.1) connected external to chamber.

4.5.4.1.1 Connect the battery and associated test equipment in accordance with 4.4.4.4.2.

4.5.4.1.2 Close the switch. From 15 seconds through 6 minutes after closing the switch the battery voltage shall be 12 plus or minus 1 volts and the output current shall be 4 plus or minus 0.33 amperes.

4.5.4.1.3 At the end of 8 plus 1 minus 0 minutes the output voltage shall be a minimum of 10.5 volts and the output current shall be a minimum of 3.5 amperes.

4.5.4.2 High temperature operation. The battery shall be placed within a test chamber, the temperature of which shall be raised to 130 degrees F plus or minus 5 degrees F and maintained at this temperature for a period of 48 hours plus or minus 5 hours. At the end of this test period the battery, while still at the test temperature of 130 degrees F, shall meet all the performance requirements and product characteristics of this specification.

4.5.5 Operating life. Life test shall consist of a series of charge and discharge cycles in accordance with Table I.

Table I

Cycle	Type	Requirements
1	Measured	3.4.1
2 - 19	Automatic	None
20	Measured	3.4.1
21 - 39	Automatic	None
40	Measured	3.4.1
41 - 59	Automatic	None
60	Measured	3.4.1
61 - 79	Automatic	None
80	Measured	3.4.1
81 - 99	Automatic	None
100	Measured	3.4.1 except that seven plus one minus zero shall be acceptable

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4.5.5.1 The first cycle shall consist of a battery charge as in 4.4.4.3 and a measured discharge as in 3.4.1.

4.5.5.2 An automatic cycle shall consist of a constant voltage charge (3.4.2.2) followed by a constant resistance discharge (using 4.3.4) for eight plus one minus zero minutes. The discharge cycle shall be followed by a rest period as described in 4.4.4.2.

4.5.5.3 If the battery fails to meet the capacity requirements of a measured cycle (Table I), as many as four repeat tests shall be permitted to demonstrate conformance to capacity requirements as specified in 3.4.1. Repeat tests shall not be counted as life cycles.

4.5.6 Standtime. The battery shall be charged in conformance with 4.4.4.3 and allowed to remain unused for a minimum period of 60 days; thereafter it shall meet the requirements of 4.4.4.

4.6 Failure of the battery to meet any of the requirements and tests of this specification shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A and Level C.

5.1.1.1 Cleaning and drying. Batteries shall be cleaned and dried by any process or combination of processes which will accomplish thorough cleaning and drying without damage to the battery.

5.1.1.2 Preservation application. None required.

5.1.1.3 Unit packaging. Each battery shall be packaged in a polyethlene bag in accordance with Specification MIL-B-22205. The bag shall be prepared from a 10 inch by 20 inch by 0.006 thick polyethlene sheet, type II, grade B, finish I (L-P-378), in accordance with BUWEPS LD 495960.

5.2 Packing

5.2.1 Number of packages per pack. Six unit packages shall be packed in each exterior carton.

5.2.2 Level A and Level C. Batteries shall be packed in accordance with BUWEPS LD 495960.

5.3 Marking

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5.3.1 Special markings. None, unless otherwise specified.

5.3.2 Normal markings. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The battery is intended to be used as a source of electrical power for Exercise Head Mk 78 Mod 0.

6.2 Ordering data

6.2.1 Procurement documents should contain the specific title, number, and date of this specification.

6.3 Upon receipt of the batteries, the electrolyte will be below the level indicated by the marking on the battery. This condition is characteristic of a nickel-cadmium battery when discharged. Refer to maintenance instructions (3.2.8) to see when additional electrolyte or distilled water is necessary.

6.4 If electrolyte is accidentally spilled from opened or broken cells onto the hands or clothing or other material, wash with cold water or a boric acid solution.

NOTICE

When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact the Government may have formulated, furnished, or in any way supplied the said drawings specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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CONTRACT NUMBER

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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